

## BPHO ROUND 0 · QUESTION PAPER

## Drill bank section 5

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking

*Original questions written for this course, in the Round 0 format. Not BPhO material.*

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**Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

**Q1** · Dimensional analysis as a construction: which combination of a density, a speed and a length has the dimensions of a mass flow rate A

Three quantities are available: a density  $\rho$  whose dimensions are  $M L^{-3}$ , a speed  $v$  whose dimensions are  $L T^{-1}$ , and a length  $d$  whose dimensions are  $L$ .

Which combination of these three has the dimensions of the mass of fluid crossing a section each second, which is  $M T^{-1}$ ?

**A**  $\rho v d^2$

**B**  $\rho v^2 d$

**C**  $\rho v / d$

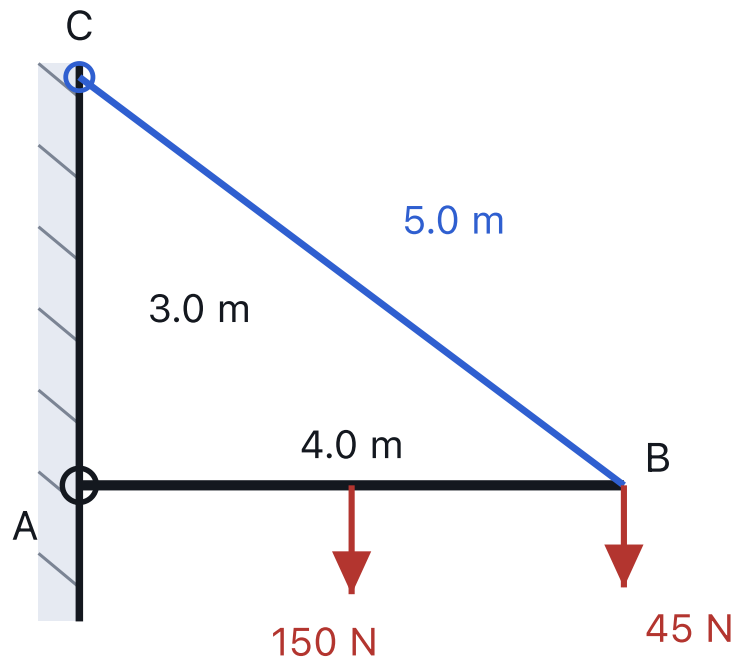
**D**  $\rho v d$

**E**  $\rho v^2 d^2$

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**Q2** · A hinged rod held by a cable: the tension from moments C

A uniform-free horizontal rod **AB** is 4.0 m long and is hinged to a wall at **A**. A cable joins **B** to a point **C** that is 3.0 m vertically above **A**; the cable is therefore 5.0 m long. A weight of 150 N acts at the middle of the rod and a load of 45 N hangs from **B**.

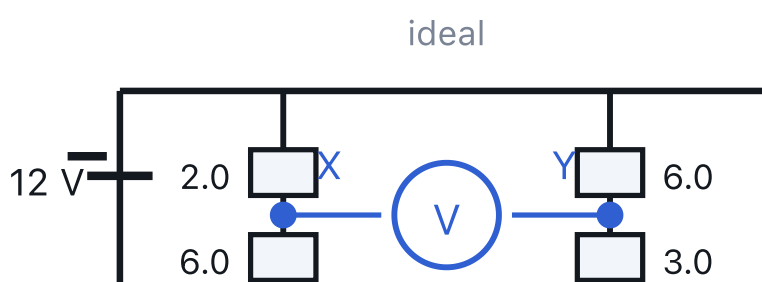


What is the tension in the cable?

- |                |                |
|----------------|----------------|
| <b>A</b> 150 N | <b>B</b> 250 N |
| <b>C</b> 200 N | <b>D</b> 120 N |
| <b>E</b> 160 N |                |
-

**Q3** · Two potential dividers: the voltmeter reading between their midpoints H

A 12 V supply feeds two branches in parallel. The left branch is 2.0 ohm above its midpoint X and 6.0 ohm below it; the right branch is 6.0 ohm above its midpoint Y and 3.0 ohm below it. An ideal voltmeter bridges X and Y.



What does the voltmeter read?

- |                |                |
|----------------|----------------|
| <b>A</b> 3.0 V | <b>B</b> 8.0 V |
| <b>C</b> 9.0 V | <b>D</b> 1.5 V |
| <b>E</b> 5.0 V |                |

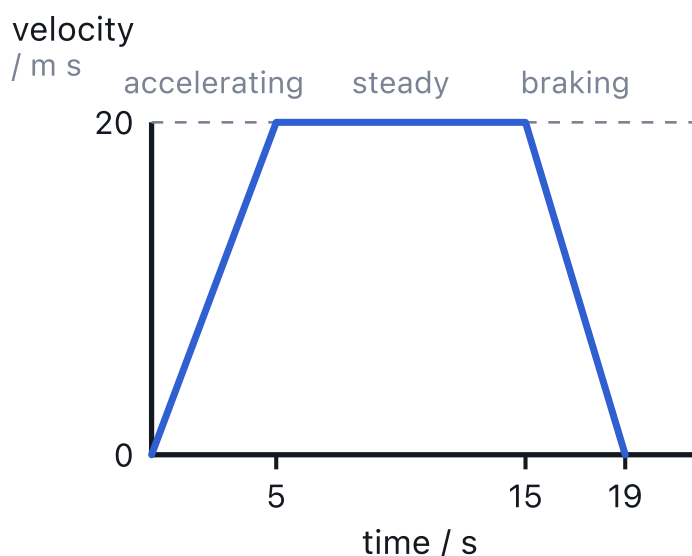
**Q4** · Order of magnitude: the number of seconds in one year A

To the nearest power of ten, how many seconds are there in one year?

- |                 |                 |
|-----------------|-----------------|
| <b>A</b> $10^5$ | <b>B</b> $10^7$ |
| <b>C</b> $10^6$ | <b>D</b> $10^8$ |
| <b>E</b> $10^9$ |                 |

**Q5** · A velocity-time graph: the total distance from the area **B**

A vehicle moves along a straight line. Its velocity-time graph is shown.

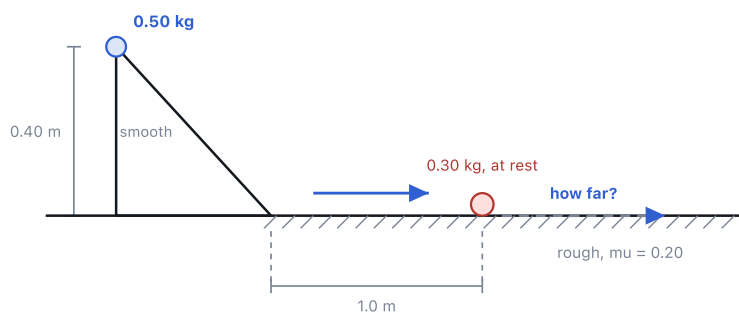


What total distance does it travel between  $t = 0$  and  $t = 19$  s?

- A** 250 m **B** 300 m  
**C** 200 m **D** 290 m  
**E** 340 m

**Q6** · A ball down a smooth ramp, across a rough floor, then sticking to a second ball: how far the combined mass slides **C**

A 0.50 kg ball is released from rest at a height of 0.40 m at the top of a smooth ramp, as the figure shows. It reaches a rough horizontal surface of coefficient of friction 0.20, slides 1.0 m along it, and then meets a 0.30 kg ball at rest. The two balls stick together and continue over the same rough surface.

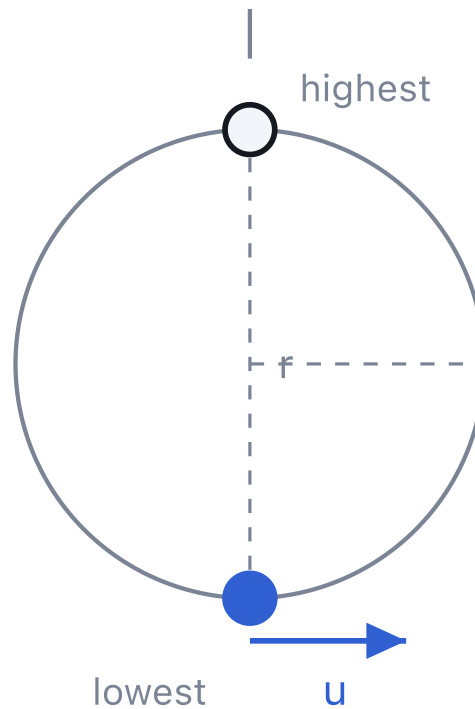


Taking  $g = 10 \text{ m/s}^2$ , how far does the combined mass slide after the collision?

- A** 0.63 m **B** 0.78 m  
**C** 0.16 m **D** 0.39 m  
**E** 0.31 m

**Q7** · A vertical circle: the minimum speed at the bottom for the string to stay taut D

A particle of mass  $m$  is tied to a light string of length  $r$  and whirled in a vertical circle. Its speed at the lowest point is  $u$ . The string must remain taut all the way round, including at the highest point, which is two radii above the lowest.

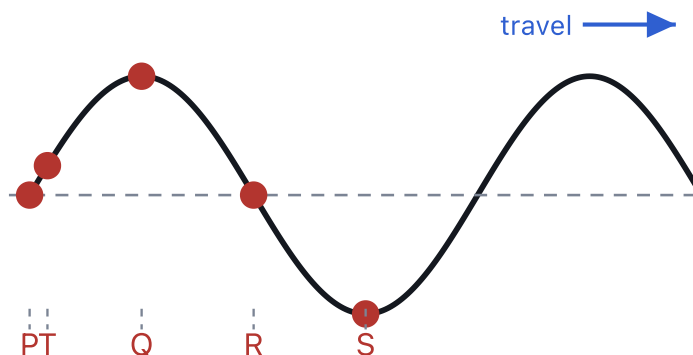


What is the minimum value of  $u$  for this to happen?

- |                                   |                                   |
|-----------------------------------|-----------------------------------|
| <b>A</b> $\sqrt{(5 \text{ g r})}$ | <b>B</b> $\sqrt{(2 \text{ g r})}$ |
| <b>C</b> $\sqrt{(3 \text{ g r})}$ | <b>D</b> $\sqrt{(g \text{ r})}$   |
| <b>E</b> $\sqrt{(4 \text{ g r})}$ |                                   |
-

**Q8** · A wave snapshot: the transverse velocity from the local slope **F**

The diagram is a snapshot of a transverse wave of amplitude 2.0 cm travelling to the right at 1.5 m/s. At the point labelled Q the string makes an angle whose slope is -0.40.

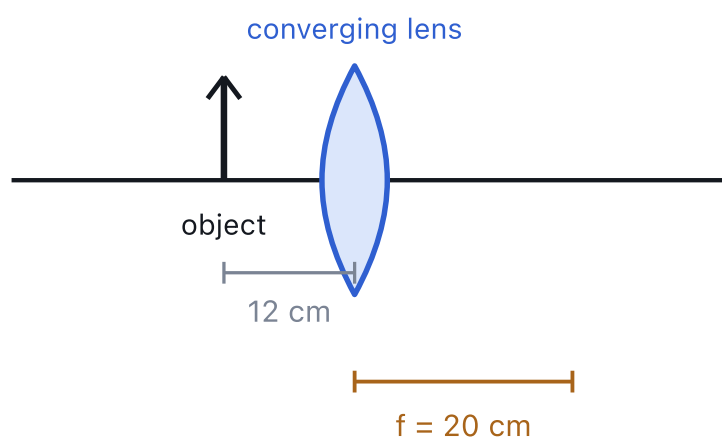


What is the transverse velocity of the point Q?

- |                            |                          |
|----------------------------|--------------------------|
| <b>A</b> 0.60 m/s downward | <b>B</b> 0.40 m/s upward |
| <b>C</b> 0.60 m/s upward   | <b>D</b> 0.90 m/s upward |
| <b>E</b> 0 m/s             |                          |

**Q9** · A converging lens: an object inside the focal length gives a virtual image **G**

An object is placed 12 cm to the left of a converging lens of focal length 20 cm.

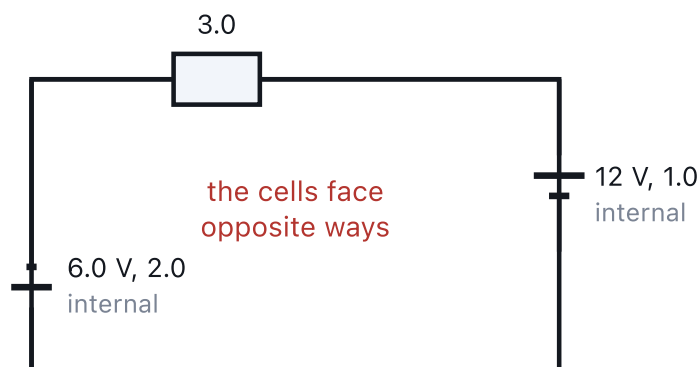


What is the linear magnification produced by the lens?

- |               |               |
|---------------|---------------|
| <b>A</b> 0.40 | <b>B</b> 1.7  |
| <b>C</b> 2.0  | <b>D</b> 0.60 |
| <b>E</b> 2.5  |               |

**Q10** · Two opposing cells: the current in the single loop H

A single loop contains a 3.0 ohm resistor, a 12 V cell of internal resistance 1.0 ohm with its positive terminal uppermost, and a 6.0 V cell of internal resistance 2.0 ohm with its positive terminal lowermost, so the two cells oppose each other.



What is the current in the circuit?

- |                                                                   |                                              |
|-------------------------------------------------------------------|----------------------------------------------|
| <p><b>A</b> 2.0 A</p> <p><b>C</b> 3.0 A</p> <p><b>E</b> 6.0 A</p> | <p><b>B</b> 1.0 A</p> <p><b>D</b> 0.50 A</p> |
|-------------------------------------------------------------------|----------------------------------------------|
- 

**Q11** · Radioactive decay: the fraction remaining after a number of half-lives K

A radioactive sample has a half-life of 8.0 days. What fraction of the original nuclei remains after 24 days?

- |                                                              |                                         |
|--------------------------------------------------------------|-----------------------------------------|
| <p><b>A</b> 1/4</p> <p><b>C</b> 1/16</p> <p><b>E</b> 1/2</p> | <p><b>B</b> 1/8</p> <p><b>D</b> 3/8</p> |
|--------------------------------------------------------------|-----------------------------------------|
- 

**Q12** · The momentum of a photon of given wavelength L

What is the momentum of a photon of wavelength 600 nm? (Use  $h = 6.6 \times 10^{-34} \text{ J s}$ .)

- |                                                                                                                                                                                                                |                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>A</b> <math>3.3 \times 10^{-19} \text{ kg m s}^{-1}</math></p> <p><b>C</b> <math>1.1 \times 10^{-19} \text{ kg m s}^{-1}</math></p> <p><b>E</b> <math>6.6 \times 10^{-34} \text{ kg m s}^{-1}</math></p> | <p><b>B</b> <math>5.5 \times 10^{-28} \text{ kg m s}^{-1}</math></p> <p><b>D</b> <math>1.1 \times 10^{-27} \text{ kg m s}^{-1}</math></p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
-

**Q13** · Converting a speed from m/s to km/h A

A sprinter covers 100 m in 10 s. What is the average speed in km/h?

**A** 36 km/h

**B** 10 km/h

**C** 3.6 km/h

**D** 360 km/h

**E** 100 km/h

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**Q14** · A horizontal projectile: the range from the time of fall B

A ball is thrown horizontally at 12 m/s from a height of 20 m above level ground. (Use  $g = 10 \text{ m/s}^2$ .) How far from the point directly below the launch does it land?

**A** 12 m

**B** 48 m

**C** 24 m

**D** 34 m

**E** 17 m

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**Q15** · Two perpendicular forces: the equilibrating force C

Two forces, 3 N and 4 N, act at right angles on a body. What single force is needed to keep the body in equilibrium?

**A** 7 N

**B** 1 N

**C** 12 N

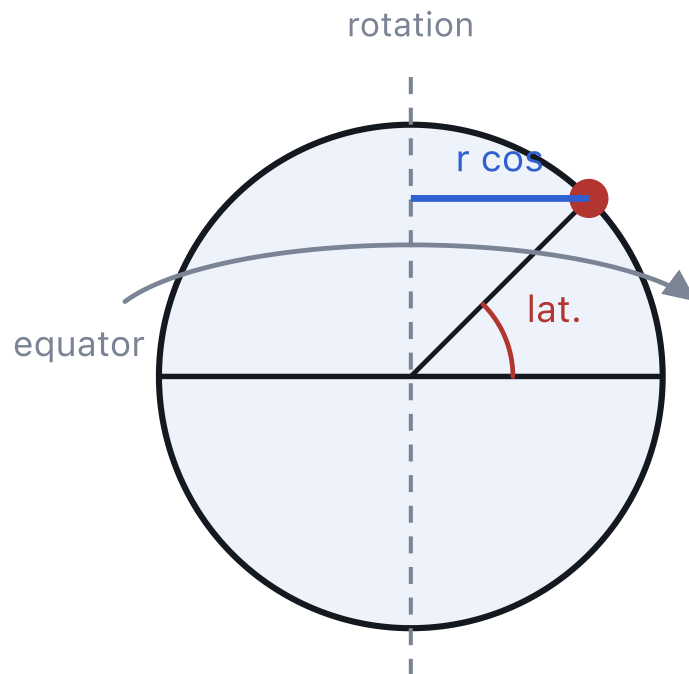
**D** 3.5 N

**E** 5 N

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**Q16** · A rotating sphere: how latitude changes the centripetal acceleration D

A sphere of radius  $R$  rotates with angular speed  $\omega$  about a vertical axis. A point at latitude  $\lambda$  moves in a circle about that axis.

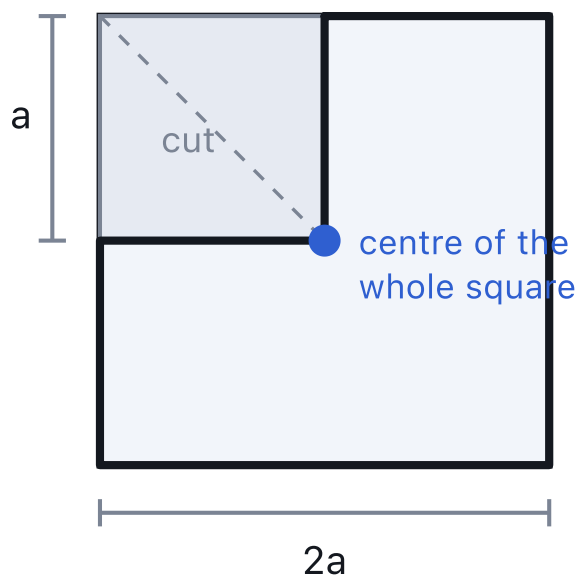


By what factor is the centripetal acceleration of that point less than the centripetal acceleration of a point on the equator?

- |                           |                           |
|---------------------------|---------------------------|
| <b>A</b> $\sin \lambda$   | <b>B</b> $\cos^2 \lambda$ |
| <b>C</b> $\sin^2 \lambda$ | <b>D</b> 1                |
| <b>E</b> $\cos \lambda$   |                           |

**Q17** · A square lamina with a corner removed: the shift of the centre of mass E

A uniform square lamina has side  $2a$ . A square of side  $a$  is removed from its upper-left corner.



By what distance is the centre of mass displaced horizontally from the centre of the original uncut square?

- |                 |                |
|-----------------|----------------|
| <b>A</b> $a/3$  | <b>B</b> $a/6$ |
| <b>C</b> $a/4$  | <b>D</b> $a/2$ |
| <b>E</b> $a/12$ |                |

**Q18** · Two coherent sources: the fringe spacing F

Two coherent sources 1.0 mm apart illuminate a screen 1.5 m away. For light of wavelength 500 nm, what is the fringe spacing on the screen?

- |                  |                  |
|------------------|------------------|
| <b>A</b> 0.25 mm | <b>B</b> 1.5 mm  |
| <b>C</b> 2.5 mm  | <b>D</b> 0.75 mm |
| <b>E</b> 0.50 mm |                  |

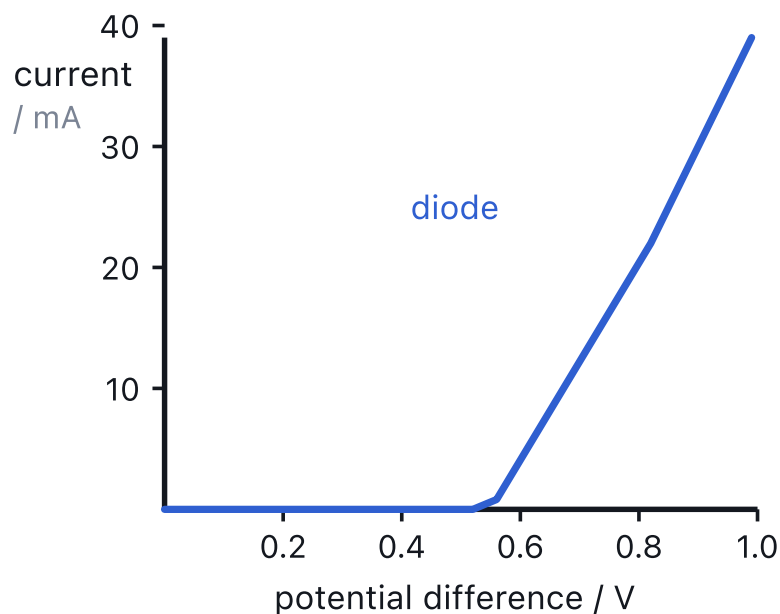
**Q19** · The critical angle: its sine for glass to air G

Light travels from glass of refractive index 1.50 to air ( $n = 1.00$ ). What is sin of the critical angle for this interface?

- |               |               |
|---------------|---------------|
| <b>A</b> 0.67 | <b>B</b> 1.50 |
| <b>C</b> 0.50 | <b>D</b> 0.33 |
| <b>E</b> 1.00 |               |

**Q20** · A diode and a load resistor: the operating current from the load line H

A diode whose characteristic is shown is connected in series with a 100 ohm resistor and a 1.0 V supply.

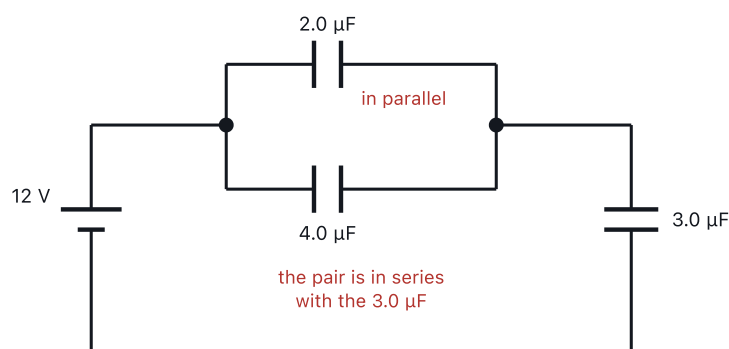


What current flows? (On the drawn characteristic the current rises linearly from 0.55 V and passes through 22 mA at 0.8 V.)

- |                 |                 |
|-----------------|-----------------|
| <b>A</b> 10 mA  | <b>B</b> 22 mA  |
| <b>C</b> 4.0 mA | <b>D</b> 2.0 mA |
| <b>E</b> 6.0 mA |                 |

**Q21** · A parallel pair in series with a third capacitor: the p.d. across the single one, from the charge the whole chain carries I

The 2.0  $\mu\text{F}$  and 4.0  $\mu\text{F}$  capacitors are connected in parallel, and that pair is joined in series with the 3.0  $\mu\text{F}$  capacitor across a 12 V supply, as the diagram shows.

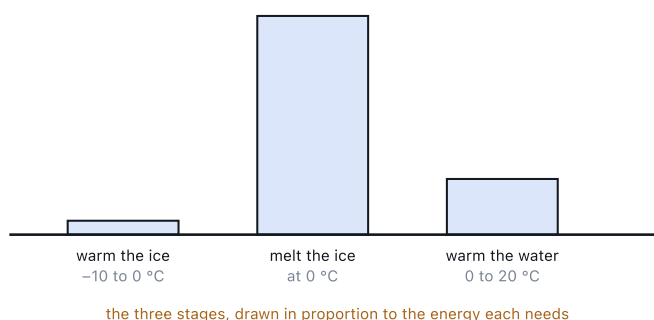


What is the potential difference across the 3.0  $\mu\text{F}$  capacitor?

- |                |                |
|----------------|----------------|
| <b>A</b> 8.0 V | <b>B</b> 4.0 V |
| <b>C</b> 6.0 V | <b>D</b> 12 V  |
| <b>E</b> 2.0 V |                |

**Q22** · Turning ice at minus ten degrees into water at twenty degrees: three stages, and the time a heater needs J

A 500 W heater supplies all of its energy to a 0.50 kg block of ice at  $-10^{\circ}\text{C}$ . The specific heat capacity of ice is  $2100\text{ J kg}^{-1}\text{ K}^{-1}$ , of water is  $4200\text{ J kg}^{-1}\text{ K}^{-1}$ , and the specific latent heat of fusion of ice is  $3.3 \times 10^5\text{ J kg}^{-1}$ . The figure shows the three stages the ice passes through, drawn in proportion to the energy each one needs.



How long does the heater take to turn the ice into water at  $20^{\circ}\text{C}$ ?

- A** 414 s
- B** 105 s
- C** 456 s
- D** 351 s
- E** 435 s

**Q23** · Mass-energy: energy released from a mass defect K

In a nuclear reaction 0.20 u of mass is converted into energy. Approximately how much energy is released? (Use  $1\text{ u c}^2 = 930\text{ MeV}$ .)

- A** 93 MeV
- B** 186 MeV
- C** 372 MeV
- D** 46 MeV
- E** 930 MeV

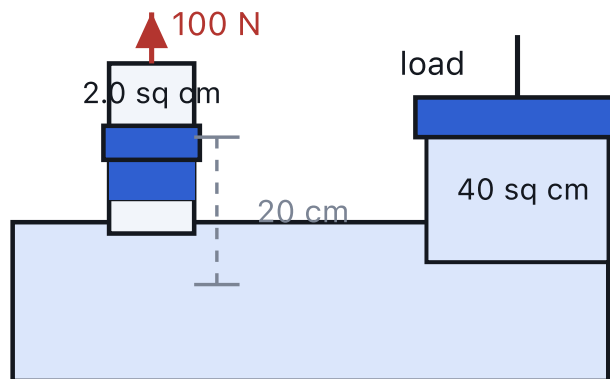
**Q24** · The photoelectric effect: maximum kinetic energy of the electrons L

Light whose photons each carry 5.0 eV of energy is incident on a metal whose work function is 2.0 eV. What is the maximum kinetic energy of the emitted electrons?

- A** 2.0 eV
- B** 5.0 eV
- C** 7.0 eV
- D** 3.0 eV
- E** 1.0 eV

**Q25** · A hydraulic jack: the load lifted by the larger piston M

A hydraulic jack has a small piston of area  $2.0 \text{ cm}^2$  pushed down with a force of  $100 \text{ N}$ , and a large piston of area  $40 \text{ cm}^2$ .



What load can the large piston support?

**A** 2000 N

**B** 100 N

**C** 400 N

**D** 50 N

**E** 500 N